Ain Shams University

Faculty of Engineering

Mechatronics Engineering Department

Collaborative Robotic Arms Digital-Twin Enabled

Graduation Project Thesis

Submitted in partial fulfillment of the requirements for the
B.Sc. degree in Mechatronics Engineering

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Declaration

We/I hereby certify that this project submitted as part of our/my partial fulfillment of BSc in Mechatronics Engineering is entirely our/my own work, that we/I have exercised reasonable care to ensure its originality, and does not to the best of our/my knowledge breach any copyrighted materials, and have not been taken from the work of others and to the extent that such work has been cited and acknowledged within the text of our/my work.

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Acknowledgment

The team would like to thank our supervisors,

Abstract

This thesis presents a Digital-Twin-enabled collaborative robotic assembly system integrating an ABB industrial manipulator and an AR4 robotic arm to execute efficient brick assembly through coordinated perception, grasping, handover, and placement. The proposed system combines global scene understanding from an Intel RealSense camera with local "eye-in-hand" sensing using an ESP32-CAM mounted on the AR4, enabling both coarse positioning and fine alignment during grasping via Image-Based Visual Servoing (IBVS). A structured state machine governs end-to-end task allocation and execution, selecting between direct assembly and cross-side handover based on brick and robot allocation logic derived from GUI/database input and global perception.

To achieve accurate interaction in a shared workspace, calibration is treated as a core requirement. AR4 calibration is performed by collecting data at multiple X–Y locations at two height levels (0 mm and 50 mm) and fitting mapping functions using multiple approaches (Radial Basis Function (RBF), linear, and polynomial). Polynomial regression demonstrated superior performance and was validated using training/testing splits. Similarly, global camera calibration based on polynomial regression is applied using samples across varying X, Y, and Z coordinates to enable reliable estimation of brick world pose across the entire table. Perception is implemented through YOLOv8 segmentation for brick and feature detection, while grasp selection is addressed using a transformer-based encoder–decoder trained with both successful grasps (True Grasps) and failure/avoidance cases (False Grasps), enabling negative reinforcement to reduce unstable grasp predictions. During fine control, homography is used to map feature points between the RealSense global view and the ESP32-CAM local view, simplifying tracking and improving approach robustness.

Experimental results demonstrate improved positional accuracy using polynomial calibration, effective brick detection performance using segmentation, increased grasp stability using the True/False labeling strategy, and robust end-to-end execution under a state-machine framework. The thesis concludes with limitations and future directions, including improved domain adaptation, force-aware handovers, and enhanced Digital Twin synchronization for higher fidelity planning and monitoring.

Contents

Chapter 1

Introduction

1.1 Background and Motivation

Collaborative robotic systems are increasingly adopted in modern automation to improve throughput, flexibility, and safety in assembly workflows. While a single industrial robot can achieve high repeatability, multi-robot collaboration enables task decomposition, parallelism, and improved workspace coverage.

Despite these advantages, multi-robot collaboration introduces substantial technical challenges. These include:

- Accurate calibration between sensing systems and robotic manipulators.
- Reliable perception under dynamic and uncertain real-world conditions.
- Robust grasp planning to avoid object slippage or collision.
- Stable closed-loop control for fine alignment during assembly.
- Coordination of system-level logic to manage roles, task handovers, and execution uncertainties.

This project addresses these challenges by implementing a two-robot collaborative assembly system, integrating an ABB industrial robot and an AR4 research robot to perform brick assembly tasks. The system leverages Digital Twin technology to provide visualization, monitoring, and structured integration of perception, planning, and execution modules.

1.2 Problem Statement

The primary problem addressed in this project is the design and implementation of a robust, two-robot collaborative assembly system capable of:

1. Detecting and localizing bricks within a shared workspace using global perception.
2. Selecting stable grasps and avoiding unstable configurations via learned models.
3. Executing safe and accurate grasping using hybrid control strategies (point control combined with image-based visual servoing).

4. Performing reliable handovers from the AR4 to the ABB robot when bricks need to cross workspace sides.
5. Maintaining a consistent world coordinate system across the workspace through calibration of cameras and robots.
6. Coordinating the full workflow via a state machine with well-defined transitions and recovery logic.

1.3 Project Objectives

The objectives of this thesis are summarized as follows:

- Develop a Digital Twin-enabled architecture integrating perception, decision logic, and robot execution.
- Achieve accurate AR4 position correction using data-driven calibration techniques (tested: radial basis function, linear, polynomial; best performance: polynomial regression).
- Calibrate the global camera-to-world mapping for reliable brick pose estimation across the entire workspace.
- Implement robust object detection using YOLOv8 segmentation for bricks and workspace features.
- Implement grasp detection using a transformer-based encoder-decoder trained with stable and unstable grasps to improve reliability.
- Implement fine alignment for AR4 grasping using image-based visual servoing aided by homography mapping between global and local camera views.
- Develop a state machine to govern setup, task allocation, grasping, handover, and assembly operations reliably.

1.4 Project Scope

This thesis focuses on the perception-to-action integration for collaborative assembly using two robotic arms and two camera systems (global and eye-in-hand). The scope includes:

- System calibration and camera-to-robot alignment.
- Object detection and grasp pose estimation.
- Visual servoing and hybrid control for precise manipulation.
- Handover logic between robots.
- Digital Twin integration for monitoring and validation.

The thesis does **not** include:

- Detailed mechanical design of the robots.
- Proprietary low-level controller programming of the ABB robot.
- Safety certification processes beyond safe experimental operation.

1.5 Key Contributions

The main contributions of this project are:

1. Implementation of a Digital Twin-enabled multi-robot assembly system for visualization, validation, and monitoring.
2. Development of a polynomial-based calibration method for precise AR4 positioning and global workspace mapping.
3. Integration of YOLOv8-based object detection for brick and feature localization.
4. Implementation of a transformer-based grasp detection network for improved grasp stability.
5. Hybrid control strategy combining point control and image-based visual servoing for fine alignment.
6. Reliable handover and task coordination using a state machine with recovery mechanisms.

1.6 Thesis Organization

This thesis is organized as follows:

- **Chapter 2** reviews literature on collaborative robotics, multi-robot handover, visual servoing, grasp detection, and Digital Twin integration.
- **Chapter 3** describes the system architecture, including calibration, perception, grasping, control, and workflow management.
- **Chapter 4** presents the experimental implementation, evaluation of calibration accuracy, perception performance, grasp prediction quality, and workflow reliability.
- **Chapter 5** concludes the thesis, summarizes key findings, and discusses potential future enhancements.

Chapter 2

System Design and Methodology

2.1 AR4 Restoration and Hardware Modifications

The AR4 robotic manipulator was handed over to the current team by the previous graduation project team after their completion of the project. At the time of handover, the robot was delivered in a fully assembled condition and was generally operational. The unit is an AR4 MK3 model designed by Chris Annin. However, due to the duration and intensity of prior usage during development and testing, several components exhibited wear and assembly-related issues. As a result, multiple mechanical and electrical corrective actions were required before the robot could be considered reliable for accurate motion execution and experimental work.

2.1.1 Joint 3 Mechanical Transmission Upgrade

Joint 3 of the AR4 robot exhibited significant backlash due to wear and limitations in the original mechanical transmission design. To address this issue, the joint transmission system was completely replaced in accordance with the official upgrade provided by Chris Annin. The upgrade consisted of a redesigned gear set and a new timing belt, both intended to improve torque transmission and reduce mechanical play within the joint.

figures/AR4 modifications/Joint3_mods.png

Figure 2.1: Upgraded mechanical transmission of Joint 3

After implementing the upgraded components, the backlash in Joint 3 was reduced from approximately 6° to 3°. This reduction resulted in smoother joint motion and improved positioning accuracy, directly enhancing the overall kinematic performance of the robot.

2.1.2 Shaft Key Re-Manufacturing in Joint 3

Despite the improvements achieved through the mechanical transmission upgrade, residual backlash was still observed in Joint 3. Further investigation revealed that the shaft key responsible for transmitting motion from the second gear stage to the robotic arm link exhibited wear and improper fit. This condition introduced mechanical play within the joint, negatively affecting positioning accuracy.

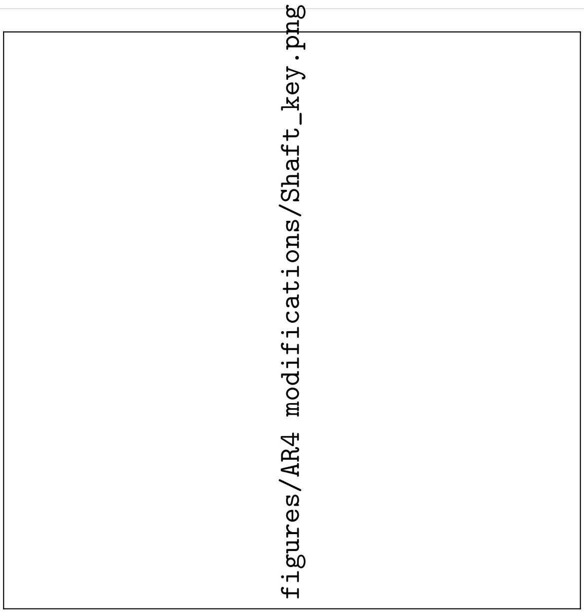

Figure 2.2: Re-manufactured shaft key used in Joint 3 to eliminate mechanical backlash

To resolve this issue, the shaft key was re-manufactured with precise dimensions to ensure a tight and accurate fit between the gear and the arm shaft using hot-wire machining. The newly manufactured key eliminated relative motion between the connected components, effectively removing the remaining backlash in Joint 3. As a result, Joint 3 achieved backlash-free motion, significantly improving the repeatability and accuracy of the robot during operation.

2.1.3 Encoder Signal Noise in Joints 5 and 6

During system testing and motion execution, unstable and noisy encoder readings were observed in Joints 5 and 6 of the AR4 robot. These fluctuations resulted in inaccurate joint position feedback, which significantly affected the forward and inverse kinematics calculations. Consequently, commanded joint positions did not always correspond to the expected physical motion of the robot.

A systematic debugging process was conducted, including signal tracing and verification of the encoder wiring and power connections. This investigation revealed that the encoders of Joints 5 and 6 were not properly connected to the regulated 5 V power supply. The improper power connection caused voltage instability, leading to fluctuating encoder signals and unreliable position measurements.

After correcting the power supply connections and ensuring proper grounding, the encoder signals became stable and consistent. This correction restored accurate joint position feedback, significantly improving the reliability of the kinematic calculations and overall motion control performance of the robot.

2.2 Cartesian Calibration of the AR4 Collaborative Robot

Accurate positioning is critical in collaborative robotic applications, particularly when multiple manipulators operate within a shared workspace. During initial experimental trials, the AR4 robotic arm exhibited noticeable Cartesian positioning inaccuracies when executing commanded end-effector positions relative to the ABB robot world coordinate frame. These inaccuracies motivated the implementation of a dedicated Cartesian calibration procedure to improve the absolute positioning accuracy of the AR4 robot.

2.2.1 Experimental Setup

The calibration experiment was conducted using a pointed end-effector mounted on the AR4 robot to enable precise position referencing. A calibrated measurement grid with millimeter resolution was placed on the worktable. The grid origin was aligned with the ABB robot world frame origin $(0, 0, 0)$, ensuring a common global reference for both manipulators.

The AR4 robot was commanded to move to a predefined set of Cartesian positions distributed over the XY plane at multiple height levels along the Z-axis. At each commanded position $(x_{cmd}, y_{cmd}, z_{cmd})$, the actual end-effector position $(x_{meas}, y_{meas}, z_{meas})$ was manually recorded using the grid.

2.2.2 Cartesian Error Characterization

The raw Cartesian positioning errors were computed as:

$$\begin{aligned} e_x &= x_{meas} - x_{cmd} \\ e_y &= y_{meas} - y_{cmd} \\ e_z &= z_{meas} - z_{cmd} \end{aligned} \tag{2.1}$$

Initial analysis revealed systematic and spatially correlated errors across the workspace, indicating that the inaccuracies were not purely random but instead dependent on the commanded Cartesian position.

2.2.3 Calibration Method Selection

Several calibration approaches were evaluated prior to selecting the final method. Table **??** summarizes the comparison between candidate techniques. The calibration dataset was divided into independent training and testing subsets to ensure that the reported performance reflects generalization rather than overfitting to the calibration points.

Method	Accuracy	Extrapolation	Data Requirements	Real-Time
Lookup Table	Medium	None	High	High
RBF Network	High	Poor	Very High	Medium
Spline Fitting	High	Limited	Medium	High
Polynomial Regression	High	Good (Low-Order)	Medium	High

Table 2.1: Comparison of Cartesian calibration methods for robotic manipulators

Based on this comparison, the polynomial calibration was selected as a balanced solution, offering a favorable trade-off between accuracy, robustness, and implementation simplicity, making it particularly suitable for real-time correction in collaborative robotic systems.

2.2.4 Polynomial Calibration Model

A second-order polynomial regression model was used to approximate the Cartesian error as a function of the commanded XY position:

$$e(x, y) = a_1 x + a_2 y + a_3 x^2 + a_4 y^2 + a_5 xy + a_6 \tag{2.2}$$

The estimated polynomial error models were used to correct the commanded positions prior to execution:

$$\begin{aligned} x_{corr} &= x_{cmd} + \hat{e}_x(x, y) \\ y_{corr} &= y_{cmd} + \hat{e}_y(x, y) \\ z_{corr} &= z_{cmd} + \hat{e}_z(x, y) \end{aligned} \tag{2.3}$$

This correction strategy enables feedforward compensation of systematic positioning errors without modifying the internal robot controller.

2.2.5 Calibration Results

Based on the experimental results, all evaluated calibration approaches significantly reduced the Cartesian positioning error when compared to the uncalibrated model. Among the tested methods, the linear model exhibited the highest residual error due to its inability to capture nonlinear kinematic and geometric effects. Radial Basis Function (RBF) networks achieved the lowest mean positioning error, demonstrating superior interpolation accuracy within the calibrated workspace. However, their locally adaptive nature resulted in less predictable behavior near the workspace boundaries and outside the training region.

The second-degree polynomial model produced slightly higher errors than the RBF-based approach, yet these errors remained within acceptable limits for the intended application. In contrast to RBF and spline-based methods, the polynomial model exhibited smoother and more stable behavior near workspace boundaries, while maintaining low computational cost and consistent real-time performance.

Figure ?? illustrates a comparison between the commanded, measured, and polynomial-corrected end-effector positions in the XY plane. The deviation between the commanded and measured positions highlights the presence of systematic Cartesian positioning errors in the AR4 robot, while the corrected positions demonstrate the effectiveness of the polynomial calibration in compensating for these errors.

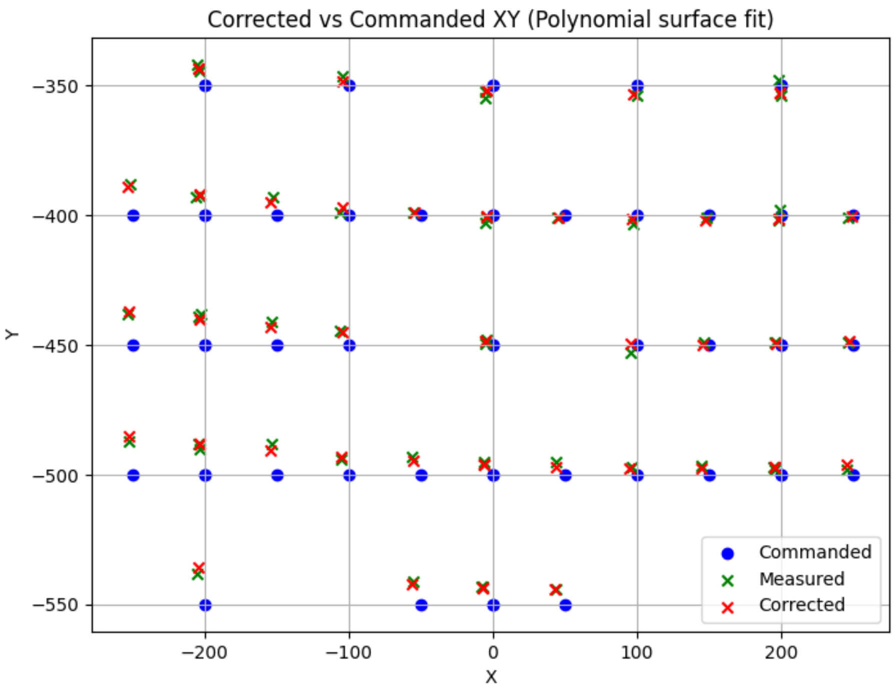

Figure 2.3: XY-plane commanded, measured, and corrected positions.

2.2.6 Error Surface Analysis

To further analyze the spatial behavior of the calibration model, three-dimensional polynomial error surfaces were generated for each Cartesian axis, which are shown in Figure ??

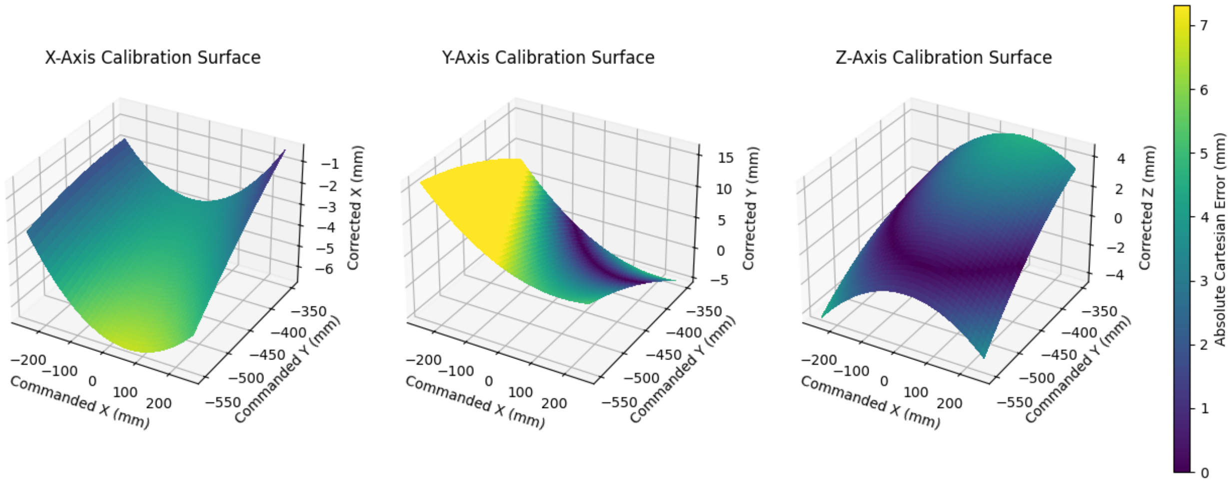

Figure 2.4: Calibration surfaces for the X, Y, and Z axes with error magnitude.

Figure ?? presents a comparison between the uncalibrated and calibrated surfaces.The comparison demonstrates that the calibration model effectively smooths and compensates for the dominant systematic error components, resulting in a more uniform and stable error distribution across the workspace.

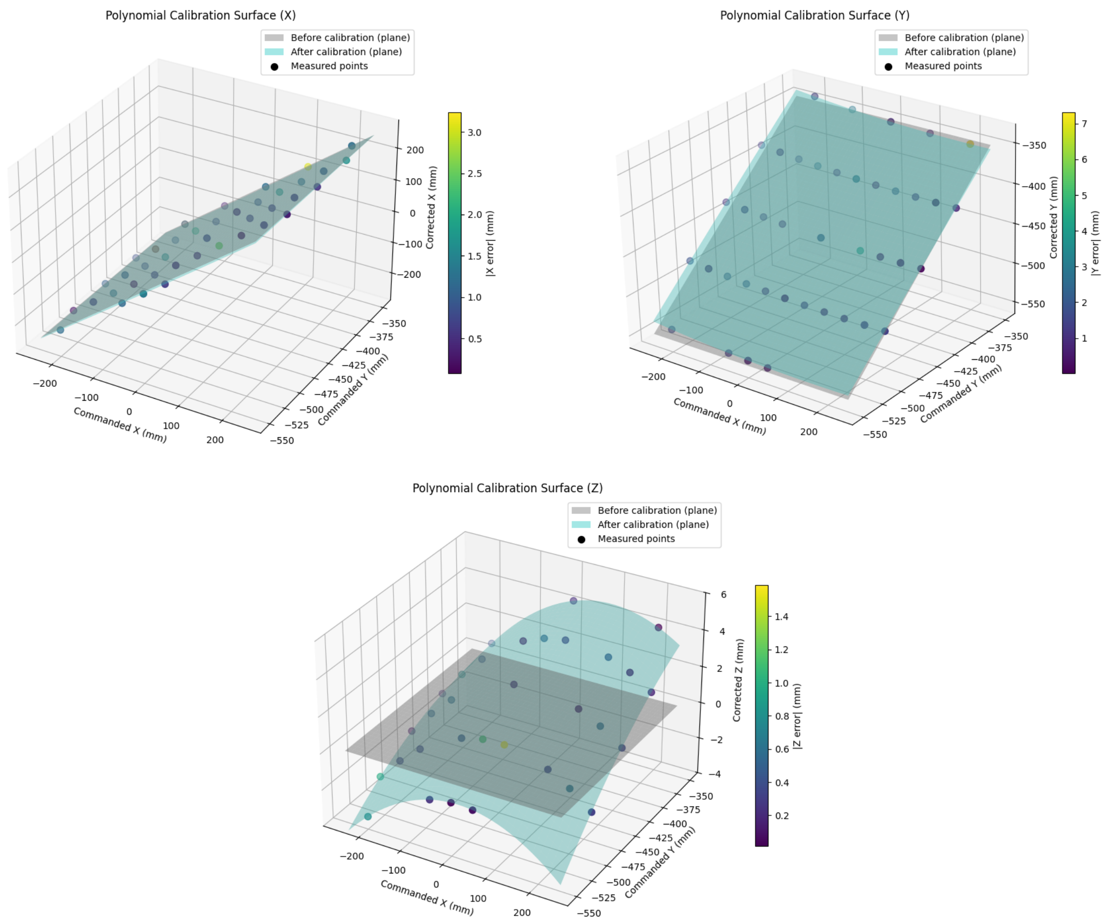

Figure 2.5: Comparison of uncalibrated and calibrated surfaces.

2.2.7 Quantitative Evaluation

The effectiveness of the calibration was quantitatively evaluated using the mean absolute error (MAE) before and after correction for each Cartesian axis, computed on the independent test dataset.

Table 2.2: Mean Absolute Cartesian Errors Before and After Calibration

Axis	**MAE Before (mm)**	**MAE After (mm)**
X	3.96	0.94
Y	5.13	1.32
Z	1.94	0.46

The quantitative results in Table **??** are consistent with the smooth and continuous behavior observed in the polynomial error surfaces shown in Figure **??**, confirming the effectiveness and stability of the proposed calibration approach.